Python Data Structures

by University of Michigan

About this Course

This course will introduce the core data structures of the Python programming language. We will move past the basics of procedural programming and explore how we can use the Python built-in data structures such as lists, dictionaries, and tuples to perform increasingly complex data analysis. This course will cover Chapters 6-10 of the textbook “Python for Everybody”. This course covers… Python 3.

Taught by:

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Basic Info	Course 2 of 5 in the Python for Everybody Specialization
Level	Beginner
Commitment	2-4 hours/week
Language	English Subtitles: Arabic, French, Bengali, Uzbek, Ukrainian, Chinese (Simplified), Greek, Italian, Portuguese (Brazil), Vietnamese, Dutch, Korean, Oriya, German, Pashto, Urdu, Russian, Thai, Indonesian, Swedish, Turkish, Azerbaijani, Spanish, Dari, Hindi, Japanese, Kazakh, Hungarian, Polish
How To Pass	Pass all graded assignments to complete the course.
User Ratings	★ 4.9

Syllabus

Module 1

Chapter Six: Strings

In this class, we pick up where we left off in the previous class, starting in Chapter 6 of the textbook and covering Strings and moving into data structures. The second week of this class is dedicated to getting Python installed if you want to actually run the applications on your desktop or laptop. If you choose not to install Python, you can just skip to the third week and get a head start.

7 videos, 7 readings

1. Video: Video Welcome - Dr. Chuck
2. Reading: Reading: Welcome to Python Data Structures
3. Reading: Help Us Learn More About You!
4. Reading: Course Syllabus
5. Reading: Textbook
6. Reading: Submitting Assignments
7. Video: Strings
8. Video: Manipulating Strings
9. Reading: Notice for Auditing Learners: Assignment Submission
10. Video: Worked Exercise: Strings
11. Video: Bonus: Office Hours New York City
12. Video: Bonus: Monash Museum of Computing History
13. Video: Fun: The Textbook Authors Meet @PyCon
14. Reading: Audio Versions of All Lectures

Graded: Quiz: Chapter 6
Graded: Assignment: Parse Text Strings

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Module 2

Installing and Using Python

In this module you will set things up so you can write Python programs. We do not require installation of Python for this class. You can write and test Python programs in the browser using the "Python Code Playground" in this lesson. Please read the "Using Python in this Class" material for details.

5 videos, 2 readings

1. Reading: Important: Using Python in this Class
2. Reading: Choosing a Text Editor
3. Video: Demonstration: Using the Python Playground
4. App Item: Python Code Playground
5. Video: Windows 10: Installing Python and Writing A Program
6. Video: Windows: Taking Screen Shots
7. Video: Macintosh: Using Python and Writing A Program
8. Video: Macintosh: Taking Screen Shots
9. Peer Review: Optional- Installing Python Screen Shots

Module 3

Chapter Seven: Files

Up to now, we have been working with data that is read from the user or data in constants. But real programs process much larger amounts of data by reading and writing files on the secondary storage on your computer. In this chapter we start to write our first programs that read, scan, and process real data.

5 videos, 1 reading

1. Video: Files
2. Video: Processing Files
3. Video: Worked Exercise: Files
4. Reading: Where is the worked exercise for the second assignment?
5. Video: Bonus: Office Hours Barcelona
6. Video: Bonus: Gordon Bell - Building Blocks of Computing

Graded: Quiz: Chapter 7
Graded: Assignment: Read a File and Convert Its Text
Graded: Assignment: Read a File and Process Its Data

Module 4

Chapter Eight: Lists

As we want to solve more complex problems in Python, we need more powerful variables. Up to now we have been using simple variables to store numbers or strings where we have a single value in a variable. Starting with lists we will store many values in a single variable using an indexing scheme to store, organize, and retrieve different values from within a single variable. We call these multi-valued variables "collections" or "data structures".

7 videos

1. Video: Lists
2. Video: Manipulating Lists
3. Video: Lists and Strings
4. Video: Fun: Python Lists in Paris
5. Video: Worked Exercise: Lists
6. Video: Bonus: Office Hours - Chicago
7. Video: Bonus: Rasmus Lerdorf - Inventing the PHP Language

Graded: Quiz: Chapter 8
Graded: Assignment: Create a Sorted Word List
Graded: Assignment: Build a List of Senders

Module 5

Chapter Nine: Dictionaries

The Python dictionary is one of its most powerful data structures. Instead of representing values in a linear list, dictionaries store data as key / value pairs. Using key / value pairs gives us a simple in-memory "database" in a single Python variable.

7 videos

1. Video: Dictionaries
2. Video: Counting with Dictionaries
3. Video: Dictionaries and Files
4. Video: Worked Exercise: Dictionaries
5. Video: Bonus: Office Hours - Amsterdam
6. Video: Bonus: Brendan Eich - Inventing Javascript
7. Video: Fun: Dr. Chuck Goes Motocross Racing

Graded: Quiz: Chapter 9
Graded: Assignment: Find the Most Frequent Sender

Module 6

Chapter Ten: Tuples

Tuples are our third and final basic Python data structure. Tuples are a simple version of lists. We often use tuples in conjunction with dictionaries to accomplish multi-step tasks like sorting or looping through all of the data in a dictionary.

6 videos

1. Video: Tuples
2. Video: Worked Exercise: Tuples and Sorting
3. Video: Bonus: Office Hours - Puebla, Mexico
4. Video: Bonus: John Resig - Inventing JQuery
5. Video: Douglas Crockford: JavaScript Object Notation (JSON)
6. Video: Fun: The Greatest Taco in the World

Graded: Quiz: Chapter 10
Graded: Assignment: Sort and Count Messages

Module 7

Graduation

To celebrate your making it to the halfway point in our Python for Everybody Specialization, we welcome you to attend our online graduation ceremony. It is not very long, and it features a Commencement speaker and very short commencement speech.

2 videos, 2 readings

1. Video: Graduation Ceremony
2. Video: Dr.Chuck Wrap Up/What's Next
3. Reading: Please Rate this Course on Class-Central
4. Reading: Post-Course Survey

How It Works

General

What do start dates and end dates mean?
Once you enroll, you’ll have access to all videos, readings, quizzes, and programming assignments (if applicable). If you choose to explore the content without purchasing, you may not be able to access certain assignments. If you don’t finish all graded assignments before the end of the session, you can reset your deadlines. Your progress will be saved and you’ll be able to pick up where you left off.

What are due dates? Is there a penalty for submitting my work after a due date?
Within a course, there are suggested due dates to help you manage your schedule and keep work from piling up. Quizzes and programming assignments can be submitted late without consequence. However, it is possible that you won't receive a grade if you submit your peer-graded assignment too late because classmates usually review assignment within three days of the assignment deadline.

Can I re-attempt an assignment?
Yes. If you want to improve your grade, you can always try again. If you’re re-attempting a peer-graded assignment, re-submit your work as soon as you can to make sure there’s enough time for your classmates to review your work. In some cases you may need to wait before re-submitting a programming assignment or quiz. We encourage you to review learning material during this delay.

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Peer-graded assignments

Peer-graded assignments require you and your classmates to grade each other’s work.

How do peer graded assignments work?
After you submit your assignment, you will review some of your peers’ assignments. The number of assignments you must review is set by the instructor of the course.

I reviewed my peers’ assignments! What happens next?
While you’re reviewing your peers’ assignments, they’ll review yours. If you submit your assignment on time, you’ll get your grade within a week, as long as at least one peer reviews your assignment. If you submit late, you’ll need all of the peer reviews the instructor requires. Learn more about Peer Graded Assignments.

How are grades calculated?
You and your classmates will be asked to provide a score for each part of the assignment. Final grades are calculated by combining the median scores you received for each section.

What kind of feedback should I give?
Use the instructor’s criteria in the rubric to grade honestly and fairly. If your peers’ answers are excellent, score them highly and tell them what they did well. If their answers aren’t as good, give the score they deserve, and be sure to provide respectful, useful feedback so they can do better next time they attempt the assignment.

Is there a penalty for submitting my work late?
No, but it’s important to submit your work as close to the due date as you can. Classmates grade most of the assignments within three days of the due date. If you submit yours too late, there may not be anyone to review your work.

If I fail an assignment, can I try again?
Yes! You can always try again, but you’ll need to resubmit your work as soon as possible to make sure your classmates have enough time to grade your work.

Can I edit my assignment?
Yes, but you’ll need to re-submit your work and any grade you’ve already received will be deleted.

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